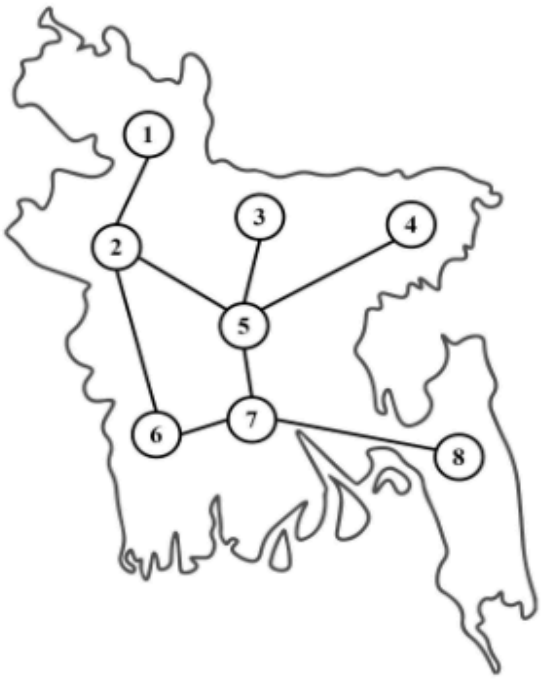
Graph

1.

The "Road Transport and Highways Division" of Bangladesh is working on a project where they want to establish six-lane highways among the 8 divisions of Bangladesh. Currently, there are 8 highways connecting these 8 divisions as given in the map below.



MAP OF BANGLADESH

1	**Rangpur**	**5**	**Dhaka**
2	**Rajshahi**	**6**	**Khulna**
3	**Mymensingh**	**7**	**Barisal**
4	**Sylhet**	**8**	**Chattogram**

Fig 1: Graph For Question 4

i. **Create** a graph representation for the given map so that it does not take more than $O(|V| + |E|)$ space/memory.

ii. **Determine** how many additional highways we need to make sure that every division is directly connected with each other.

iii. In a meeting, the director looked at the map and wondered if the divisions of Bangladesh could be split into two groups (**A** and **B**) such that all highways directly connect locations in Group A to Group B, with no highways between locations within the same group. Using an appropriate data structure, simulate the algorithm that can determine if such a division is possible.

iv. Based on the given graph in Fig 1, **explain** how you can find out if there is an odd number cycle present in the graph. An odd number cycle is a cycle that has an odd number of vertices.

2.

The cities in Italy are connected in such a way that their adjacency list representation looks as follows-
Rome: Milan, Venice, Florence
Milan: Rome, Florence, Bologna, Venice
Florence: Milan, Naples, Rome, Bologna, Venice
Venice: Rome, Milan, Florence
Naples: Florence, Bologna, Genoa
Turin: Genoa
Bologna: Naples, Genoa, Milan, Florence
Genoa: Turin, Bologna, Naples
Palermo:Catania, Syracuse
Catania: Palermo, Syracuse
Syracuse: Catania, Palermo

Suppose Rome suddenly gets affected with covid-19. The rest of the cities are non-affected. You are asked to find out the time steps the covid virus will take to affect all the non-affected cities if it takes one time step to travel from one city to another. So, one affected city will affect all of its neighbors in one time step.

a) **Draw** the graph and create the adjacency matrix for the graph. **04**
b) **Mention** the name of the graph traversal algorithm you think is suitable for solving the problem. **01**
c) **Simulate** the suitable algorithm to find the time steps needed for each non affected city to be affected. Choose the smallest (in alphabetical order) vertex when there is a choice. **05**

3.

a. **CO1** You have the following adjacency matrix for a graph. However, some of the entries are missing. Your job is to find these missing entries with the help of some clues. Then draw the graph. **06**

	A	B	C	D	E	F
A	0	0		0	0	0
B	1	0	1	0	0	1
C	0	0	0	0	1	0
D	0	0	1	0	0	0
E	0		0	0	0	
F	1	0	1	0		0

Clues:
- B can not be reached from C
- A can be reached from D
- 2 x |E| = 3 x |V|
  (twice the no. of edges is equal to thrice the no. of vertices)
- In-degree of C is not a prime number.

b. **CO2** Is an adjacency list more memory efficient than an adjacency matrix? Justify your reasoning with respect to directed vs undirected, weighted vs unweighted, sparse vs dense graphs. **04**

4.

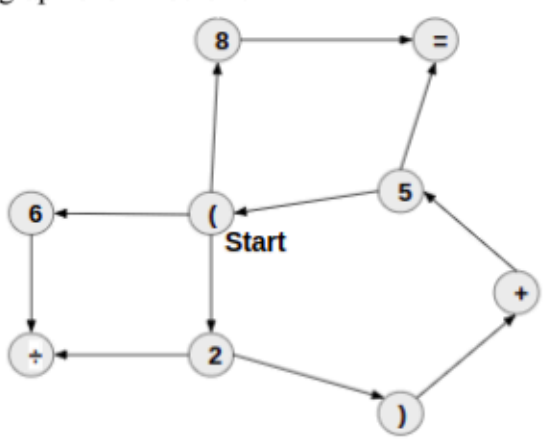
Bill has this weird characteristic of playing with different types of graphs. This time he comes up with an undirected simple graph of 10 nodes. Nodes are labeled from 3 to 12. Some of the edges are:
(5,6), (5,10), (6,11), (7,11). He created other edges following two rules.

a. All the nodes that are labeled with a number which is a multiple of 3 have edges among them.
b. All the nodes that are labeled with a number which is a multiple of 4 have edges among them.

a. CO3	**Draw** the graph to show all the edges.	**1**
b. CO3	Bill says, there are at least four triangles in the graph. Do you agree? **Support** your answer by showing the nodes which form these triangles.	**2**
c. CO2	**Simulate** a BFS algorithm on the graph to find the shortest distance from Node 3 to all others nodes.	**6**
d. CO3	Bill told you to keep on adding edges between the nodes according to your wish, keeping the graph simple (without adding multiple edges between any two nodes, self edges). **Compute** the number of more edges that you can add.	**1**

5.

Consider the graph shown below:



Bill says he has found a valid mathematical equation while running DFS from the node denoted by '('. According to him, the equation is:

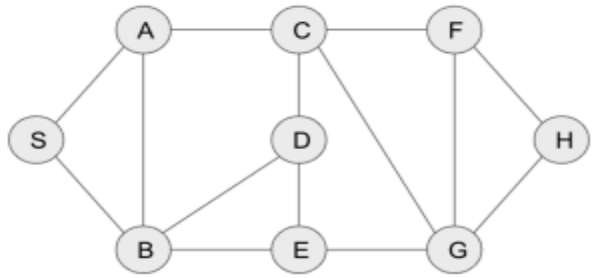
$$(6 \div 2) + 5 = 8$$

a. CO3	**Compute** the number of edges this DFS tree (of the equation) contains.
b. CO3	Is what Bill said right i.e is the equation achievable by running such a DFS? **Validate** it by showing steps. (Neighbor/Edge selection should be done according to the necessity of the verification process)
c. CO2	**Classify** the edges of the graph into tree edge, back edge and forward edge while running DFS from the node denoted by '('.

6.

Total Marks: 10**[CO3]**

Consider this graph:



a. **Justify** $\sum_{v \in V} deg(v) = 2m$, where m = number of edges in the graph. 2

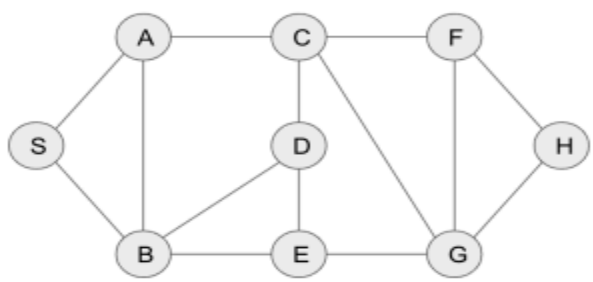
b. **Create** an adjacency matrix for this graph. 2

c. **Apply** Depth First Search to traverse this graph considering S to be the source vertex. **Demonstrate** the results. 6

7.

Total Marks:10**[CO3]**

Consider this graph:



a. **Justify** $\sum_{v \in V} deg(v) = 2m$, where m = number of edges in the graph. 2

b. **Create** an adjacency list for this graph. 2

c. **Apply** Breadth First Search to traverse this graph considering S to be the source vertex. **Demonstrate** the results. 6

8.

You are the coach of the renowned football team "Real Madrid". Your team is behind one goal and there is still some time left to back in the game by scoring a goal. Benzema is your main striker and if you can pass the ball to him he will give the goal for sure. Currently the ball is in your goalkeeper's (Courtois) hand. So, your main target should be Passing the ball in minimum steps from Courtois to Benzema as the time left is very low.

Courtois can pass the ball to Rudiger or Alaba
Rudiger can pass the ball to Modric
Alaba can pass the ball to Modric, Hazard or Nacho
Modric can pass the ball to Benzema
Hazard can pass the ball to Kroos
Nacho can pass the ball to Vini
Kroos can pass the ball to Benzema
Vini can pass the ball to Benzema

Now answer the following questions.

a) Using a suitable algorithm, **find** the minimum number of passes your team will give to reach the ball from Courtois to Benzema. **Mention** the name of the algorithm, **Show** each step of the simulation properly to find out the minimum number of passes.
b) **Find** out the players who are required for this minimum number of passes scenario.

9.

You are the coach of the renowned football team "FC Barcelona". Your team is behind one goal and there is still some time left to back in the game by scoring a goal. Lewandowski is your main striker and if you can pass the ball to him he will give the goal for sure. Currently the ball is in your goalkeeper's (Stegen) hand. So, your main target should be Passing the ball in minimum steps from Stegen to Lewandowski as the time left is very low.

Stegen can pass the ball to Alba or Roberto
Alba can pass the ball to Torres
Roberto can pass the ball to Busquets or Jong
Busquets can pass the ball to Torres or Gavi
Jong can pass the ball to Gavi
Gavi can pass the ball to Lewandowski
Torres can pass the ball to Lewandowski

Now answer the following questions.

c) Using a suitable algorithm, **find** the minimum number of passes your team will give to reach the ball from Stegen to Lewandowski. **Mention** the name of the algorithm, **Show** each step of the simulation properly to find out the minimum number of passes.
d) Find out the players who are required for this minimum number of passes scenario.

10.

4.	**a.** **CO1**	You have the following adjacency matrix for a graph. However, some of the entries are missing. Your job is to find these missing entries with the help of some clues. Then draw the graph.	**06**

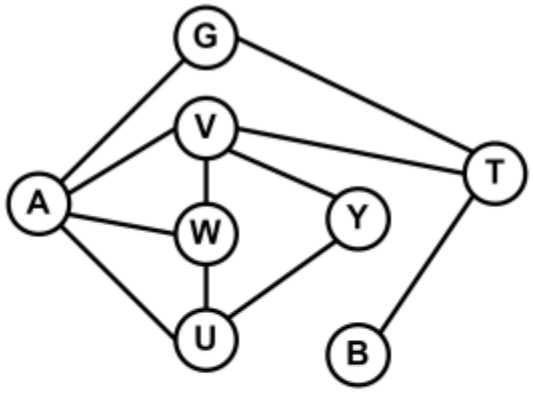
	A	B	C	D	E	F
A	0	1	0	0	0	0
B	0	0	0	0	0	1
C	0	1	0	1	0	
D	0		0	0	0	0
E	0	1	1	1	0	0
F	0	0		0		0

Clues:

- E can not be reached from B
- D can be reached from A
- 2 x |E| = 3 x |V|
  (twice the no. of edges is equal to thrice the no. of vertices)
- In-degree of B is not a prime number

	b. **CO2**	Is an adjacency list more memory efficient than an adjacency matrix? Justify your reasoning with respect to directed vs undirected, weighted vs unweighted, sparse vs dense graphs.	**04**

11.



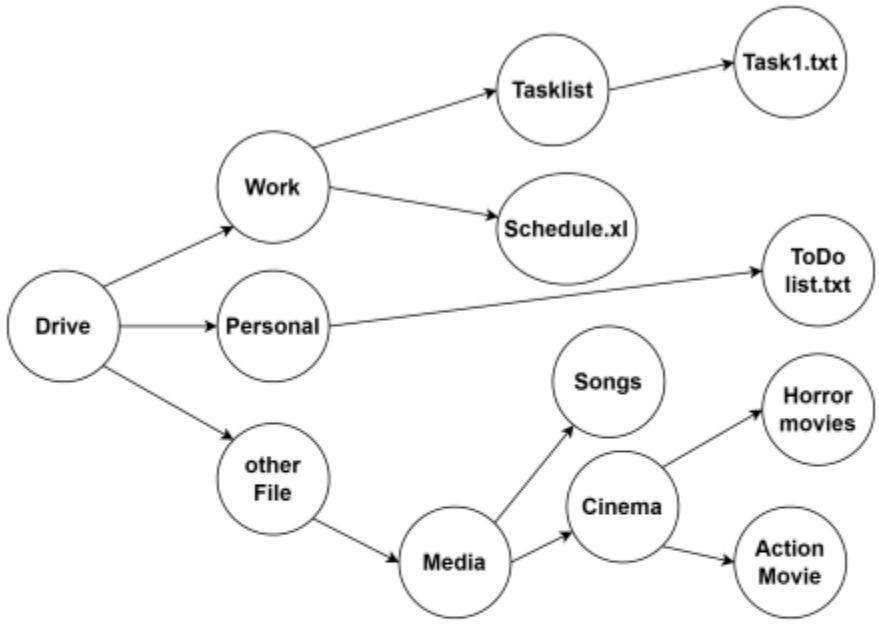
i. **Give** the adjacency matrix representation of this graph.

ii. **Determine** whether the Graph is Bipartite/Bicolorable. Show a valid grouping/coloring of the vertices.

iii. **Show** a simulation of BFS algorithm with proper use of data structure and other necessary details to find the shortest path from A to B .

12.

You are a computer science student and you are given a file structure encoded in a graph. You want to make a navigator system where if a person wants to find a file they can enter the name of the file and your program will return the exact location of the file.
For example, consider the following graph:



If the user wants to find task1.txt; your program will give the following output: Drive->Work->Tasklist->task1.txt
If the user wants to find Horror Movies; your program will give the following output:
Drive->other file->Media->Cinema->Horror Movies

Now answer the following questions:

i. **Give** the adjacency list representation of this graph. You can use either the whole name or a shorter version of each node.
ii. **Write** the name of your preferred algorithm to solve the above mentioned problem (a navigator system). Explain your reasoning in brief.
iii. **Show** a simulation of your presented solution with proper use of data structure and other necessary details to give the desired answers as the sample input shows.

13.

Dhaka, especially your area, is facing an Electricity crisis. Many initiatives including load shedding, reducing office hours, etc have already been taken. The local authority of your area is looking for feasible solutions to reduce the consumption of electricity. One of the members of their advisory committee has suggested that they should reduce the operating costs of road lighting. Till now every road is illuminated all night long. To reduce electricity consumption, they have decided to no longer illuminate every road, but to switch off the road lighting of some roads. To make sure that the inhabitants of your area still feel safe, they want to optimize the lighting in such a way that after darkening some roads at night, there will still be at least one illuminated path from every major point in your area to every other major point.

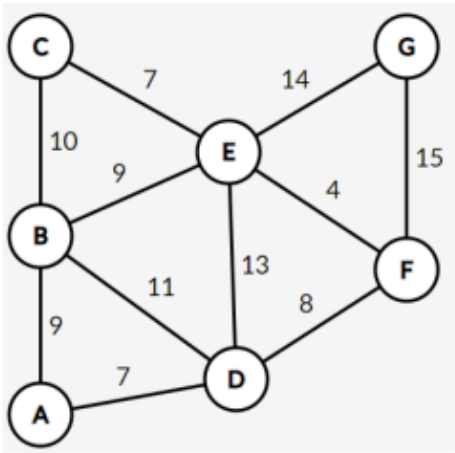


Figure : Graph of your area where the Vertices denote major points and Edges denote electricity consumption in a day.

a. CO3 **Apply** a suitable Algorithm to help the authority determine the roads they need to illuminate and the cost which will minimize the electricity consumption. If you need to consider any root vertex for your Algorithm, you can consider 'A' as such.

b. CO5 Observe the Graph of your area and your provided solution of question 1(a) and **determine** the maximum electricity of a day the authority can save.

c. CO6 The United Arab Emirates (UAE), especially Dubai, is hosting the Asia Cup 2022. They too want to minimize the electricity consumption by switching off the lights in some roads leaving exactly one illuminated path from every major point in your area to every other major point. However, to keep the tourists entertained, the Authority of Dubai has taken an opposite approach to yours. They want to illuminate the roads which require the most electricity.
Propose an algorithm to compute the path which fulfills this requirement.

14.

As we all know, the Spanish National Football Team is notorious for their passing football, especially their back passes. After their disastrous 2022 World Cup campaign, The King of Spain made you the coach and declared that no back passes are allowed on the field from now on till Spain scores 2 goals, that is, Simon the goalkeeper can only pass to the defenders Rodri and Cesar, but Rodri and Cesar cannot pass the ball back to Simon. Also, every pass now comes with a cost– and the more cost players accumulate, the angrier the King gets. These are the costs of the passes:

Passes	Cost
Simon → Rodri	2
Simon → Cesar	4
Rodri → Cesar	1
Rodri → Pedri	7
Cesar → Gavi	3
Gavi → Pedri	2
Pedri → Morata	1
Gavi → Morata	5

Find the lowest cost for the ball to reach every player starting from Simon, using any algorithm you find useful. The King will need to see every single step of the algorithm, or he will not believe you and fire you on the spot.

After scoring 2 goals, Spain changed their strategy and started to back pass again incorporating 2 more players, the master of dark arts Busquets and the young sensation Ansu Fati. Now they are no longer concerned with the cost of the pass as well. These are the passes for the new strategy:

Passes
Simon → Rodri
Simon → Cesar
Rodri → Busquets
Rodri → Pedri
Cesar → Rodri
Busquets → Gavi
Busquets → Pedri
Gavi → Cesar
Gavi → Pedri
Gavi → Fati
Fati → Morata
Pedri → Fati
Morata → Pedri

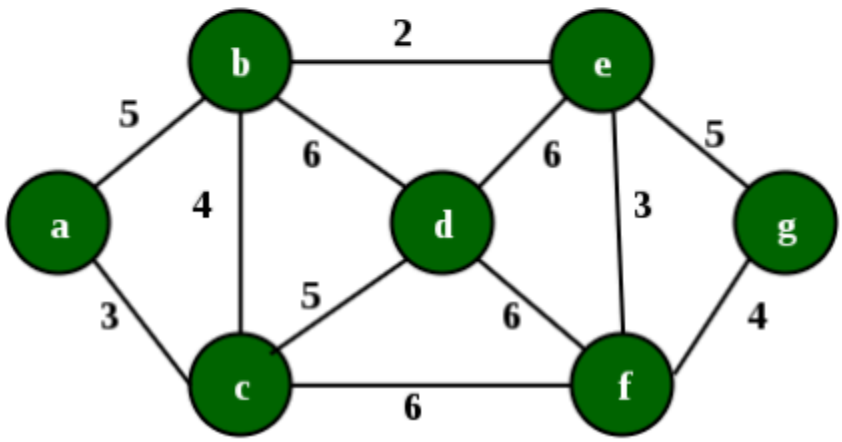
Now using a suitable algorithm, **find** out the largest group of players who can pass the ball among themselves. For example, one such group can be (Fati, Morata, Pedri) where Morata can pass to Pedri, Pedri can pass to Fati and Fati can pass to Morata. It is important to keep in mind that the king will be observing every step of the algorithm. Any discrepancies can lead you to lose your job.

15.

You work as an engineer for the Roads and Highways department in a district. The district has 7 thanas that are represented by the 7 vertices in the following graph. The edges between the vertices represent the roads that connect one thana with another and the weight of an edge represents the length of the road.

A recent flood has totally damaged these roads and immediate repair work is needed. The cost to repair a road is proportionate to its length. However, your department does not have the budget to repair all the roads so you need to repair a subset of the roads. Which algorithm should you use here to find out the roads that need to be repaired to keep all the thanas connected to one another with the **minimum** possible cost?

Show the step by step simulation of the algorithm. You may pick any algorithm of your preference that you think will solve the problem.



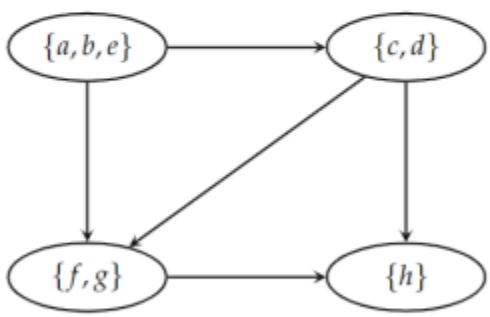
16.

Your friend is trying to find the strongly connected components of a directed graph G. Recall that this involves first running depth-first search on G, and then running depth-first search on the reverse graph, G^R, using the decreasing order of finish time. Your friend has done this and found the following table. Here, $a, b, c, d, e, f, g, h, i$ are the vertices of G.

	a	b	c	d	e	f	g	h	i
finish (G)	6	4	8	18	5	14	17	16	15
finish(G^R)	18	17	12	10	16	6	8	9	4

Now answer the following questions.

a) **Find** a vertex in G that is guaranteed to be contained in a source connected component. Recall that a source connected component in an SCC DAG is a node that has indegree zero. For example, in the following SCC DAG, $\{a, b, e\}$ is a source connected component.



This image is just an example to show how SCC can convert a graph into DAG, and the source connected component

b) Let x be your answer to (a). **Find** all the vertices in the strongly connected component that contains x.

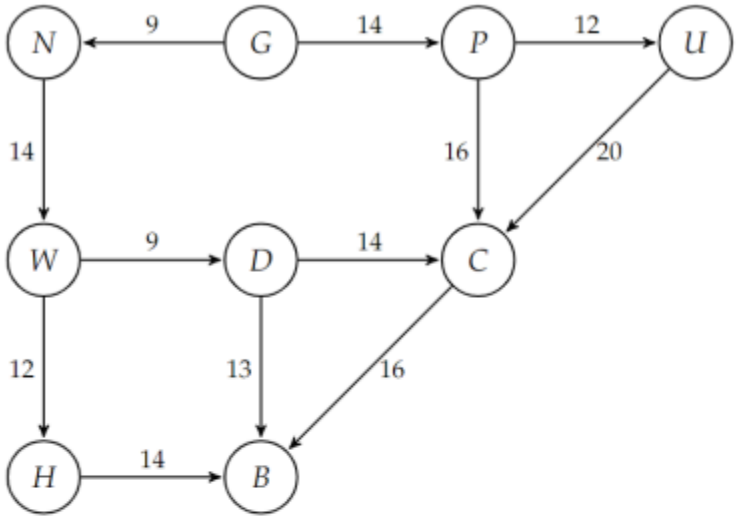
c) If the vertices you found in (b) are deleted from G, which set of vertices is guaranteed to be a source connected component of the resulting graph? **Write** the list of those vertices.

d) How many strongly connected components does G have? Just **write** the answer.

e) **Draw** the DAG of the strongly connected components of G. There is more than one correct answer, anyone of those will be accepted.

17.

Consider the network of roads connecting a set of cities given by the graph below.



a) **Simulate** Dijkstra's algorithm to find the weight of the shortest path from G to B.

b) There is a proposal to add one new road to this network, and there is a list of five pairs of cities between which the new road can be built. Each such potential road has an associated length. As a designer for the public works department, you are asked to determine the road whose addition to the existing network would result in the maximum decrease in the distance from G to B. One way to do this would be to add each of the five edges separately, run Dijkstra's algorithm five different times, and then compare the results. But can you do this by invoking Dijkstra's algorithm only two times (and possibly doing a linear amount of extra work)?

Present your idea with a pseudo-code/flowchart/step by step instruction/algorithm explanation.

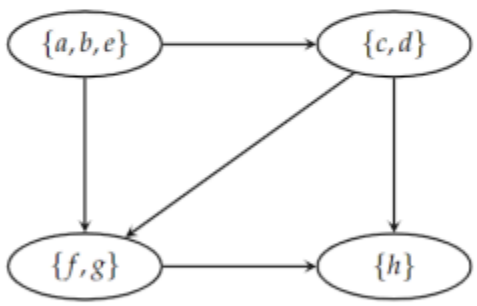
18.

Your friend is trying to find the strongly connected components of a directed graph G. Recall that this involves first running depth-first search on G, and then running depth-first search on the reverse graph, G^R, using the decreasing order of finish time. Your friend has done this and found the following table. Here, a, b, c, d, e, f, g, h, i are the vertices of G.

	a	b	c	d	e	f	g	h	i
finish(G)	7	8	9	14	6	18	13	17	16
finish(G^R)	16	18	12	6	17	10	7	8	9

Now answer the following questions.

a) **Find** a vertex in G that is guaranteed to be contained in a source connected component. Recall that a source connected component in an SCC DAG is a node that has indegree zero. For example, in the following SCC DAG, $\{a, b, e\}$ is a source connected component.



This image is just an example to show how SCC can convert a graph into DAG, and the source connected component

b) Let x be your answer to (a). **Find** all the vertices in the strongly connected component that contains x.

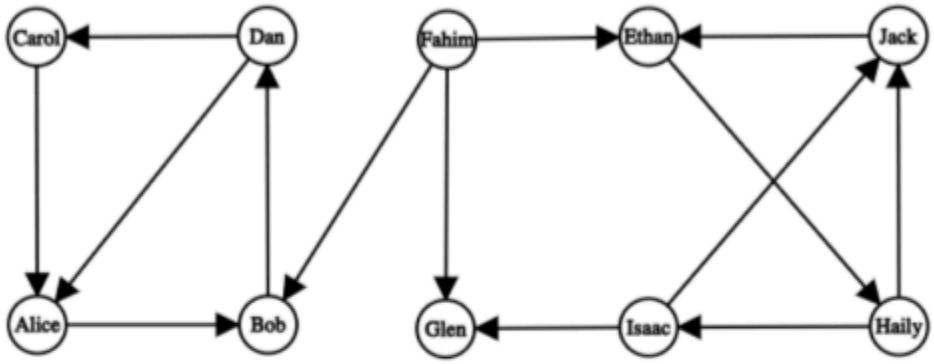
c) If the vertices you found in (b) are deleted from G, which set of vertices is guaranteed to be a source connected component of the resulting graph? **Write** the list of those vertices.

d) How many strongly connected components does G have? Just **write** the answer.

e) **Draw** the DAG of the strongly connected components of G. There is more than one correct answer, anyone of those will be accepted.

19.

Among a group of 10 friends, it's not feasible for each individual to visit the houses of all the others. With an upcoming tournament in mind, the friends need to establish groups. The criterion for forming these groups is that a friend must be capable of visiting the houses of every other member in their group. The task involves calculating the count of viable groups following this criterion. A group has to consist of a minimum of two people.



a) [CO1] The end result will showcase the list of groups and the members. **Name** the algorithm tailored for this situation.

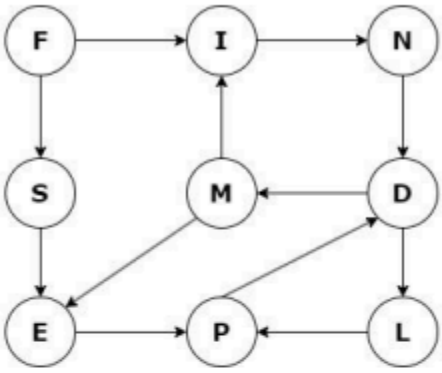
b) [CO1] **Show** a simulation of your chosen algorithm using the graph above. **Write** the number of groups that can be formed and the members of each group.

c) [CO1] Those who are incapable of forming a group will be identified as individuals ineligible for tournament participation. **Identify** those people if there are any.

d) [CO3] We also want to find the person who can visit the most houses. **Propose** an algorithm to implement this. Present your algorithm with a pseudocode/flowchart/step-by-step instructions.

20.

Sherlock Holmes found himself in a dire situation where James Moriarty imprisoned his assistant Watson in a maze. The maze is made of 9 different rooms. Each room has a part of the key with an alphabet written on it. Moriarty tells Sherlock that only if he can attach all the key chain parts in exactly this order: **"F-I-N-D-M-E-P-L-S"**, then he can unlock the room with that key where Watson is kept. The map of the maze is given below:



a) [CO1] Now you know something about a graph traversal algorithm that travels from the source to as far as possible along each branch (depth-wise traversal). Simulate/Draw the traversal and **fill up** the following table (in your script) with the start and ending time for each node you achieved while doing this task. Keep in mind, wherever you have to choose between different neighbors to explore, follow the sequence that will make the key chain parts in the sequence that Sherlock needs.

Node	D	E	F	I	M	S	L	P	N
Start									
End									

b) [CO1] When Sherlock tries to save his friend, water suddenly fills the rooms. He needs to find a way to contain the water somehow so that the water doesn't overflow from one place to another. So, he created different isolated closed groups of rooms where he can travel anywhere in that particular group, meaning they have a path from any room to any other room inside the group. After doing so, he finds out that he can make a few of these groups of rooms inside this maze. A group can contain 1 to N number of rooms where N is the total number of vertices, and no two groups can share the same room.

i) **Mention** the name of a suitable algorithm for this task.

ii) **Show** simulation for the graph given above.

You can reuse/refer to your answer from question (a) to reduce work.

21.

a) [CO2] "Dijkstra's Algorithm can find the single source shortest paths for any graph."
Do you think this statement is true? If yes, **state** your reasoning. Else, **validate** it to be false with an example graph.

b) [CO1] Imagine a daring explorer on a mission to uncover hidden treasures scattered across a cluster of islands. The listed islands have a bidirectional bridge between them:

Aqua	Blaze
Aqua	Coral
Blaze	Coral
Blaze	Dune

Coral	Dune
Blaze	Ember
Dune	Ember
Dune	Flare

Coral	Flare
Ember	Flare
Ember	Glimmer
Flare	Glimmer

If each bridge has a travel time of 5 minutes, give an O(V+E) algorithm that can find the shortest travel time from Aqua to every other island.
Explain your idea in an executable code/pseudocode/flow-chart format.

c) [CO1] Continuing on the previous question, suppose the travel time of each bridge is not equal anymore. The updated travel times are as follows:

Bridge between		**Travel time**
Aqua	Blaze	5
Aqua	Coral	13
Blaze	Coral	4
Blaze	Dune	6
Coral	Dune	5
Blaze	Ember	7

Bridge between		**Travel time**
Dune	Ember	6
Dune	Flare	6
Coral	Flare	16
Ember	Flare	3
Ember	Glimmer	11
Flare	Glimmer	2

Now, the explorer wants to find a set of bridges that connects all the islands with the **minimum total travel time**. However, the bridge between Ember and Flare and the bridge between Flare and Glimmer suddenly become vulnerable and may break down anytime. So, the explorer decides to avoid these two bridges while constructing the subgraph that connects all islands.

Answer the following questions based on this scenario.
i) Which bridges will the explorer choose?
ii) What will be the total travel time?
Show your works on the answer script.

22.

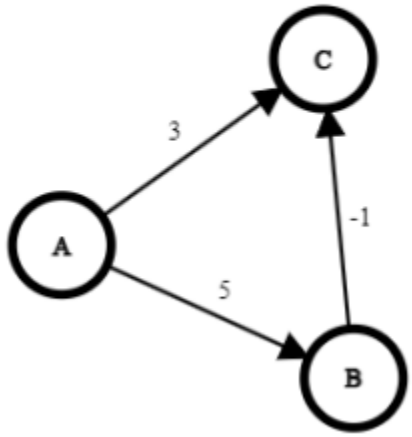
Professor Bumstead needs to figure out a proper order of clothing when getting dressed. He notes down some of the dependencies which describe which garments he needs to put on before others. For example, Prof Bumstead needs to put on his shirt before he puts on his tie or his belt. He wrote down the following:

1. undershorts, watch and socks are independent
2. pants need to be put on after undershorts
3. shoes need to be put on after pants, undershorts and socks
4. belt needs to be put on after pants and shirt
5. tie needs to be put on after shirt
6. jacket needs to be put on after tie and belt

a. **Write** down the name of an appropriate algorithm to find the proper order of clothing items.
b. **Simulate** the algorithm by drawing a proper graph and determining the correct order of clothing items. If you can choose among multiple items, choose the alphabetically smallest one. For example, between "shirt" and "shoes", i comes before o, so choose "shirt".
c. Based on the clothing dependencies provided by Professor Bumstead, **identify** if there are any strongly connected components in the graph representation using an algorithm.
Demonstrate the steps of the algorithm you are applying or justify your answer in brief.

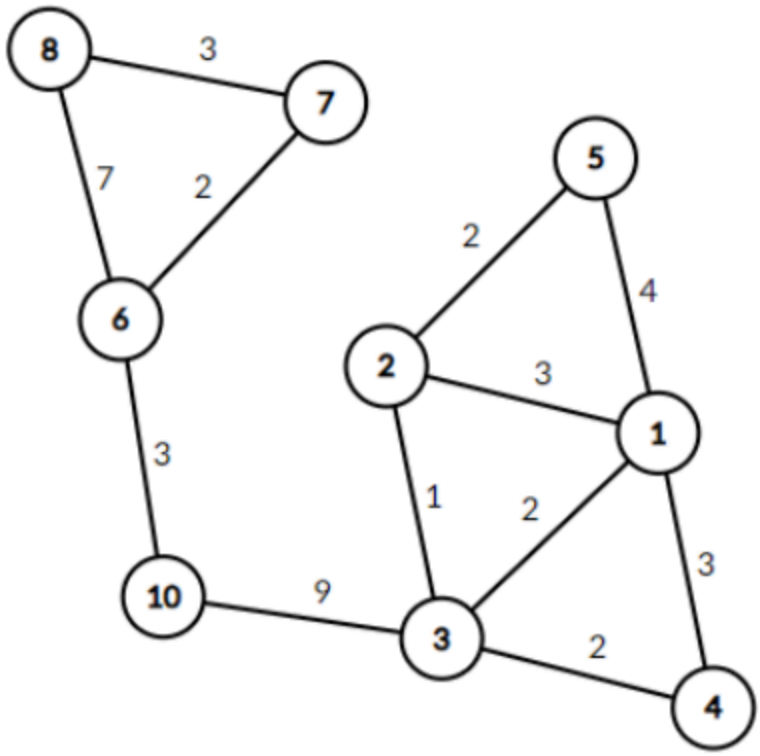
23.

You have this graph where there is a negative weight on one edge. **Explain** why Dijkstra's shortest path algorithm will find correct answers on this graph given node A as the source. Then **propose** a change in the graph to make the result incorrect.



Someone mistakenly used Max-heap data structure instead of Min-heap to create a priority queue. So a pop/remove operation on the queue brings out the highest value element instead of the lowest one. Then this priority queue was used to implement Dijkstra's algorithm.
Should it still work on any input graph? **Explain** your answer with proper reasoning and/or example. Assume that there will be no negative edges in the input.

Construct the minimum spanning tree from the graph below. Present your work with proper details/data structures. Finally, **draw** the MST and **write** the total cost of that tree.



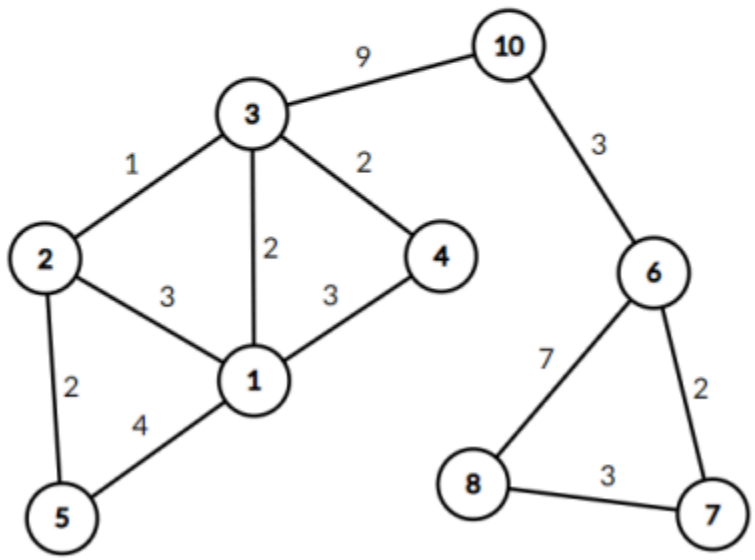
24.

Dijkstra's algorithm inserts all vertices in a priority queue/Min-heap at the beginning and then repeatedly extracts the vertex with minimum distance. However, someone proposed the following changes:

1. At the beginning, it inserts only the source vertex in the priority queue.
2. Whenever the distance of a vertex is updated, it then inserts that vertex into the queue.

Should the time complexity of the algorithm be same as before? **Explain** your answer with proper reasoning and/or example. Assume that there will be no negative edges in the input.

Construct the minimum spanning tree from the graph below. Present your work with proper details/data structures. Finally, **draw** the MST and **write** the total cost of that tree.



25.

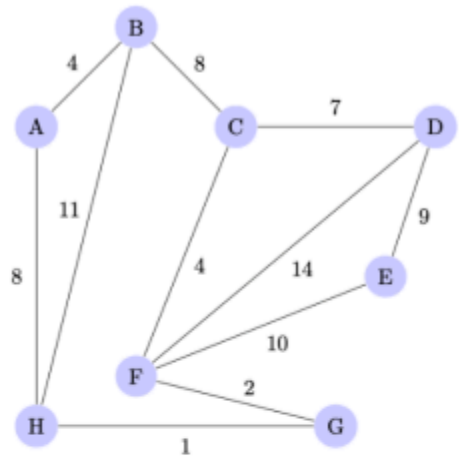


Figure 1: A Transportation Network

a. For the given transportation network, find a configuration connecting all regions with minimal cost. **Demonstrate** the steps and the algorithm you used to select connections.

b. Based on the result from part (a), **calculate** the total cost of your configuration and explain in short, why it represents the minimum cost required to connect all regions.

c. Now, imagine the road between Region C and Region F became ***unavailable*** due to maintenance. How would this impact overall connectivity? Propose an alternative solution to maintain network connectivity with minimal disruption, and explain your reasoning.

26.

You and your friend learned about single-source shortest path algorithms and are now given a directed weighted graph (*figure 2.*) to implement your algorithm. However, **the edge (2,4)** has an undefined weight **(Weight: X)**.

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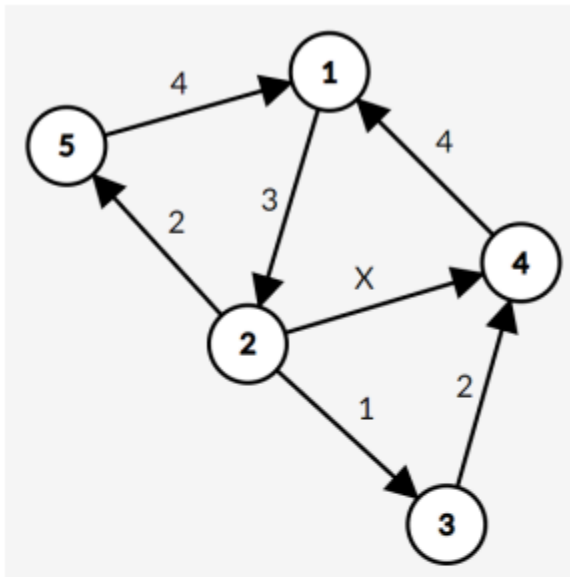

Figure 2: A Directed Weighted Graph for Question 2

If the edge weight **X = 1**, then simulate the Dijkstra Algorithm with the necessary calculation needed to find the shortest path from **Source Node 1** to every node in the graph. Mention the minimum distances you could find.

Now, with the weight of edge (2,4) set to **X = -9**, prove or disprove the statement: '*This graph's single-source shortest path is unsolvable from source 1, and only Bellman-Ford can identify this issue.*' Simulate/Give logical reasoning to support your answer.

If all roads have the same travel time, can an algorithm with time complexity O(V+E) be used to find the best route? Explain your idea in brief statements/pseudocode/flow-chart/step-by-step instructions.

27.

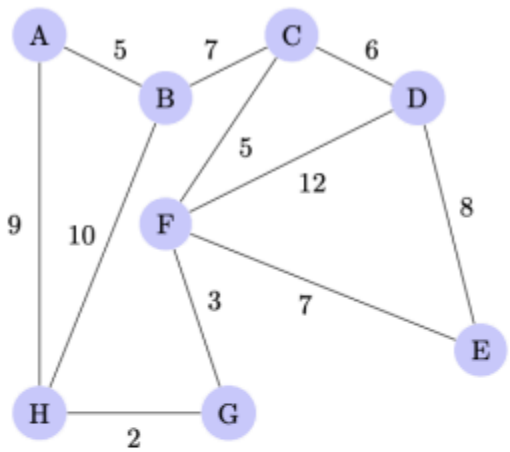

Figure 3: A Transportation Network

a. For the given transportation network, find a configuration connecting all regions with minimal cost. **Demonstrate** the steps and the algorithm you used to select connections.

b. Based on the result from part (a), **calculate** the total cost of your configuration and explain in short, why it represents the minimum cost required to connect all regions.

c. Now, imagine the road between Region C and Region D became ***unavailable*** due to maintenance. How would this impact overall connectivity? Propose an alternative solution to maintain network connectivity with minimal disruption, and explain your reasoning.

28.

You and your friend learned about single-source shortest path algorithms and are now given a directed weighted graph (*figure 2.*) to implement your algorithm. However, **the edge (2,5)** has an undefined weight **(Weight: X)**.

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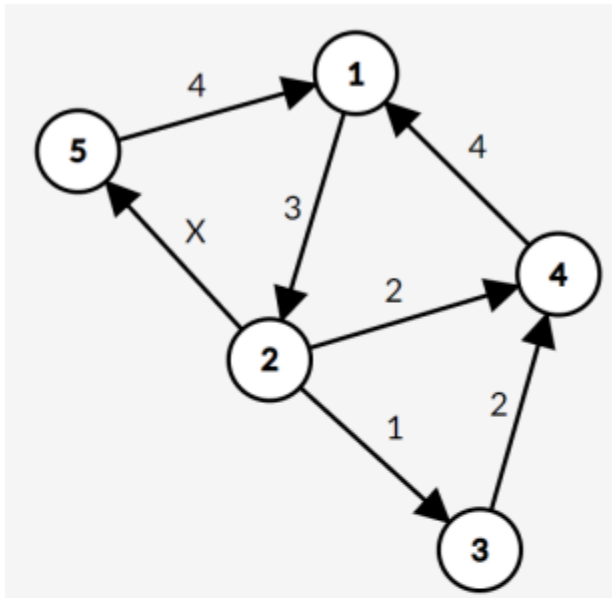


Figure 4: A Directed Weighted Graph for Question 2

If the edge weight **X = 2**, then simulate the Dijkstra Algorithm with the necessary calculation needed to find the shortest path from **Source Node 1** to every node in the graph. Mention the minimum distances you could find.

Now, with the weight of edge (2,5) set to **X = -11**, prove or disprove the statement: '*This graph's single-source shortest path is unsolvable from source 1, and only Bellman-Ford can identify this issue.*' Simulate/Give logical reasoning to support your answer.

If all roads have the same travel time, can an algorithm with time complexity O(V+E) be used to find the best route? Explain your idea in brief statements/pseudocode/flow-chart/step-by-step instructions.

29.

You have just joined a new university and are exploring the campus, which consists of several buildings connected by one-way bridges. These bridges form a directed graph, where each node represents a building and each edge represents a bridge between two buildings. You aim to analyze how the buildings and bridges are connected to understand the campus layout better and improve your navigation between classes and facilities.

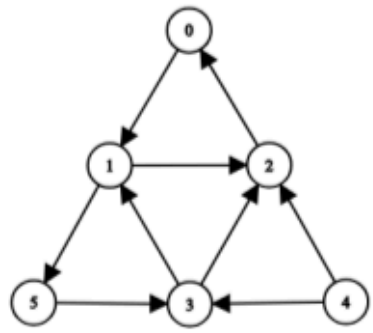


Figure 1

CO2 a) **Mention** which graph representation will let you check whether any two buildings have a bridge between them in O(1) time.

b) From Figure 1

CO1 i. **Simulate** an algorithm to find the Strongly Connected Components (SCC).

CO3 ii. Suppose a bridge gets blocked. **Explain** how you can efficiently calculate the updated SCCs from b(i) without rerunning the algorithm on the entire graph.

CO3 c) A building is called "*Accessible*" if it's possible to start from there, visit some (or all) of the other buildings, and eventually return to that building.

Propose an algorithm with a pseudocode/programmable code/step-by-step logical instructions to determine whether every building on the campus is *Accessible*. The algorithm should run in **at most O(V × (V + E))** time.

30.

Imagine, you and your best friend are trapped in a dungeon, trying to escape quickly. The dungeon is made up of chambers connected by two-way tunnels. However, these tunnels may contain dangers that drain your energy fighting it. You have a map showing how much energy you will require while passing through each of the tunnels. The energy requirements for each tunnel are not necessarily the same.

Now answer the following questions.

a. **CO1** You are given the map as an Adjacency List where the chambers are numbered from 1 to N. You are in chamber 1 and aiming to reach chamber N using a path that requires minimum energy.

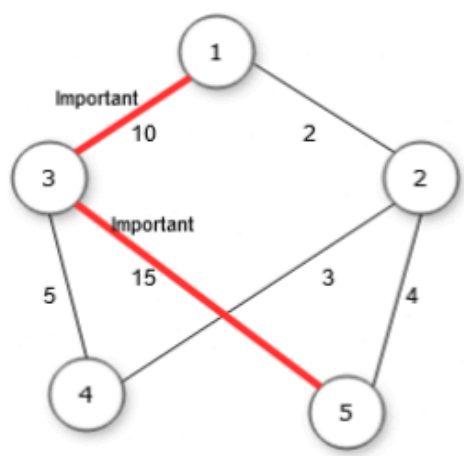
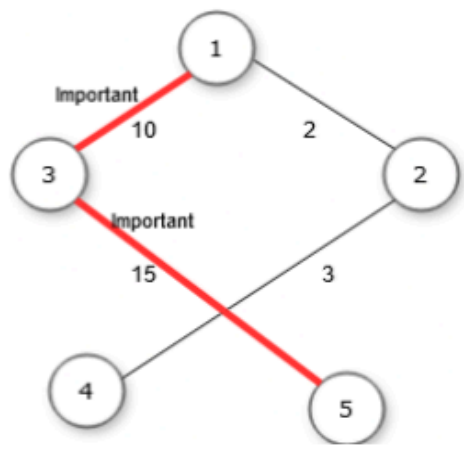
Name the algorithm you prefer to solve this problem. **Explain** your choice.

b. **CO2** If some tunnels contain an energy drink that will *restore* your energy by a certain amount, will your proposed algorithm in (a) work? **Explain** Briefly.

c. **CO2** You realize that you forgot some of your belongings in some tunnels. Hence, you want to construct the MST such that those tunnels are <u>always</u> included no matter how dangerous they are.

Formally, you have an undirected weighted graph with some edges marked as "*important*" and you need to propose a modification to Kruskal's algorithm such that you can find the MST that always includes those important edges (even if it increases the total cost).

For simplicity, you can assume the important edges/tunnels do not create any cycles. Look at the example below.

Input Graph	Modified Minimum Spanning Tree
	

Present your algorithm with a programmable code/ pseudocode/ step-by-step instruction.

d. **CO3** The map of the island is too big to carry. Thus, your friend came up with an idea to use Kruskal's Algorithm to only store the Minimum Spanning Tree (MST) of the graph. He stated that "*The minimum energy needed to travel between any two chambers in the MST is the same as in the Original Map*".

Do you agree with the statement? **Justify** your answer with proper examples and necessary details.

31.

You have just joined a new university and are exploring the campus, which consists of several buildings connected by one-way bridges. These bridges form a directed graph, where each node represents a building and each edge represents a bridge between two buildings. You aim to analyze how the buildings and bridges are connected to understand the campus layout better and improve your navigation between classes and facilities.

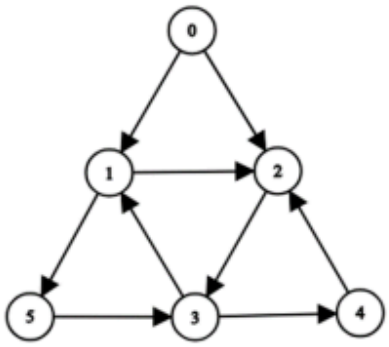


Figure 1

CO2 a) **Mention** which graph representation will let you check whether any two buildings have a bridge between them in O(1) time.

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CO3 c) A building is called "*Accessible*" if it's possible to start from there, visit some (or all) of the other buildings, and eventually return to that building.

Propose an algorithm with a pseudocode/programmable code/step-by-step logical instructions to determine whether every building on the campus is *Accessible*. The algorithm should run in **at most O(V × (V + E))** time.

32.

Captain Jack Sparrow has arrived at a treasure island, planning to steal the treasure quickly. The island has two-way pathways, each with hidden dangers. To handle these dangers, Jack brought a weapon. Facing danger in a pathway requires using some bullets. He has a map of the island showing the number of bullets needed to fight the dangers in the pathways. Bullet requirements for each pathway are not necessarily the same.

Now, answer the following questions.

a. CO1 Jack has been given the map as an Adjacency List where the locations are numbered from 1 to N. Jack is in chamber 1 and aiming to reach location N using a path that requires minimum bullets.

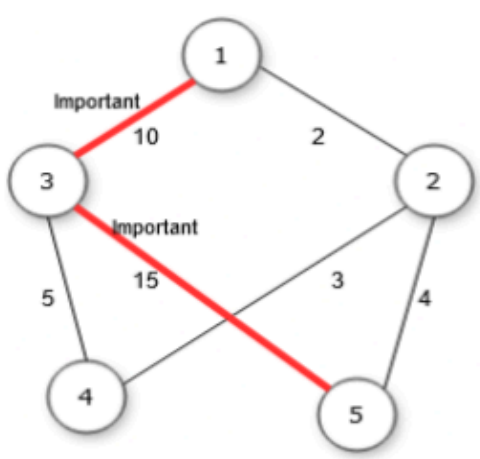
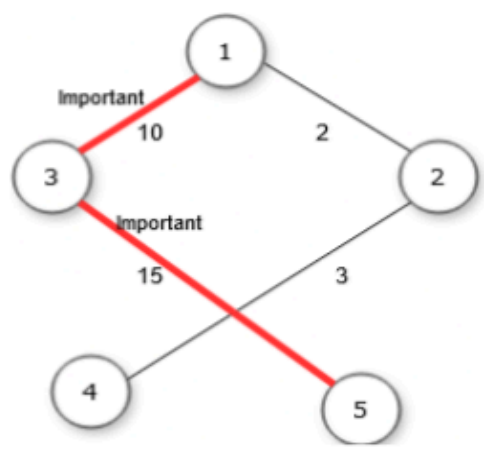
Name the algorithm you prefer to solve this problem. **Explain** your choice.

b. CO2 If some pathways contain hidden supplies of bullets that will refill his ammo by a certain amount, will your proposed algorithm in (a) work? **Explain** Briefly.

c. CO2 Jack realizes that he left some of his belongings in some pathways. Hence, he wants to construct the MST such that those tunnels are <u>always</u> included no matter how dangerous they are.

Formally, he has an undirected weighted graph with some edges marked as "important" and he needs to propose a modification to Kruskal's algorithm such that he can find the MST that always includes those important edges (even if it increases the total cost).

For simplicity, you can assume the important edges/tunnels do not create any cycles. Look at the example below.

Input Graph	Modified Minimum Spanning Tree
	

Present your algorithm with a programmable code/ pseudocode/ step-by-step instruction.

d. **CO3** The map of the island is too big to carry. Thus, your friend came up with an idea to use Kruskal's Algorithm to only store the Minimum Spanning Tree (MST) of the graph. He stated that "*The minimum energy needed to travel between any two chambers in the MST is the same as in the Original Map*".

Do you agree with the statement? **Justify** your answer with proper examples and necessary details.

33.

a. **CO1** For a directed graph, does the number of back edges equal the number of cycles present? **Explain** briefly. **01**

b. **CO1** Given a directed graph G = (V, E) represented with an adjacency list, **explain** how you can find whether the graph is a directed acyclic graph (DAG) or not. Your method must be as asymptotically efficient as possible. **01**

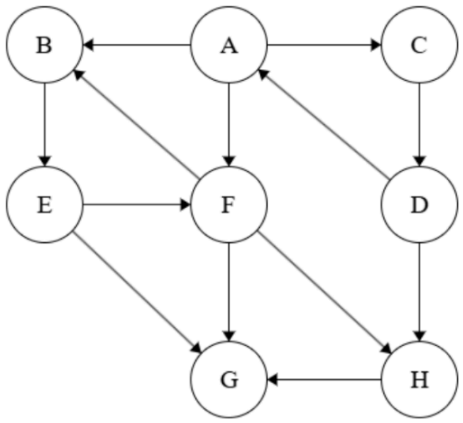


Figure 1: Q1(b)

c. **CO1** **Simulate** Kosaraju's algorithm on the given graph Q1(b) to find the strongly connected components (SCCs). Label each SCC with the set of nodes it contains. While performing graph traversal, use node A as the starting node and break ties in alphabetical order. **04**

d. **CO1** Now, treat each SCC identified in Q1(c) as a single "super node." In this new graph, every edge from the original graph either becomes internal (and thus disappears, since it stays within an SCC) or becomes an edge connecting two different SCCs. This transformation results in a Directed Acyclic Graph (DAG). **Draw** this DAG. **01**

e. **CO1** Given a set of SCCs, what is the maximum number of new SCCs that can appear if we delete one edge from the DAG of these SCCs? **01**

Consider the following algorithm:

- Step 1: Perform DFS on the given graph. For each node, keep track of the finishing time (the time when the algorithm finishes exploring all descendants of the node).
- Step 2: Sort the nodes in ascending order of the finish time.
- Step 3: Perform DFS in the sorted order of the nodes. For each DFS traversal, include each reachable node as one SCC.

f. CO2 **Prove** that the given algorithm is incorrect by providing a graph as a counterexample, on which the algorithm will fail to find the correct set of SCCs. **02**

35.

2 The country of Technovia is setting up an emergency grid that connects 8 major cities: A (Capital), B, C, D, E, F, G, H. These cities are connected by one-way roads. Each road has a traffic weight which is used to estimate the delivery delay due to congestion. Medical aid and personnel must be delivered from the capital (A) to all other cities. The map of the country is given below:

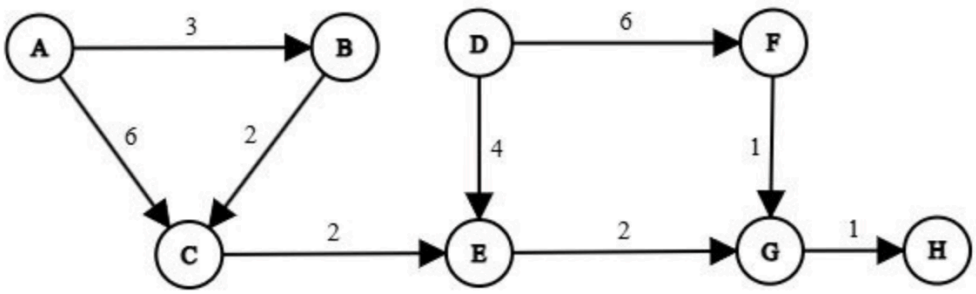


Figure 2: Q2.

a. CO1 Suppose you, an algorithm enthusiast, have been assigned to determine the minimum delivery time from capital A to all other cities. **04**

1. Which SSSP (single source shortest path) algorithm would you prefer to solve the problem?
2. **Simulate** your mentioned algorithm with the necessary data structures to find the minimum delivery times from the capital to all other cities. Travel times are equal to the traffic weights of the roads.

b. CO3 Continuing from the given scenario, the Ministry of Transport enforces that while delivering medical resources, after using any high-traffic road (traffic weight > 4), the next road used in the delivery path must have a traffic weight ≤ 4. This is to prevent back-to-back gridlock. **02**

Propose a modification to your algorithm of Q2(a) to determine the minimum delivery time from capital A to all other cities while strictly following the additional constraint of traffic weights. **Present** your idea using programmable code/pseudocode/step-by-step instructions.

c. **CO1**	Now the Ministry wants to design a minimum-traffic infrastructure plan that connects all the cities (an MST of the graph). Will Kruskal's or Prim's MST finding algorithm work properly for the given one-way roads? If yes, **justify** your answer. Otherwise, **propose** necessary modifications to the graph so that the algorithm works.	**02**
d. **CO3**	While finding the MST of the graph, we want to enforce a rule that '*No city can have more than two (02) roads*'. Propose a modification to the MST algorithm (either Kruskal's or Prim's) to accommodate this requirement. **Present** your idea using programmable code/pseudocode/step-by-step instructions.	**02**